

Homework Set 4, CPSC 6430/4430

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K -NN and Decision Boundary

The table below provides a training dataset containing six observations, three predictors and one qualitative response variable.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

Suppose we wish to use this data set to make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K -nearest neighbors (K -NN).

1. Compute the Euclidean distance between each observation and the test point $X_1 = X_2 = X_3 = 0$.
2. What is our prediction with $K = 1$ and why?
3. What is our prediction with $K = 3$ and why? Please determine the probability assigning to each class (Red or Green) by using uniform weight or distance weight (where each neighbor's weight is determined by $w_i = \frac{\exp\{-d_i^2\}}{\sum_{j=1}^K \exp\{-d_j^2\}}$ and d_i denotes the Euclidean distance of testing data to i -th nearest neighbor data) respectively.
4. Now if we only use the first two features X_1 and X_2 , please scatter plot the six observation points and plot the contour for decision boundary by referring to here.

Solution

Let the test point be

$$x_0 = (0, 0, 0).$$

For each observation $x_i = (X_{i1}, X_{i2}, X_{i3})$, the Euclidean distance to x_0 is

$$d_i = \sqrt{(X_{i1} - 0)^2 + (X_{i2} - 0)^2 + (X_{i3} - 0)^2}.$$

1. **Euclidean distances from each observation to $(0, 0, 0)$:**

$$d_1 = \sqrt{0^2 + 3^2 + 0^2} = \sqrt{9} = 3,$$

$$d_2 = \sqrt{2^2 + 0^2 + 0^2} = \sqrt{4} = 2,$$

$$d_3 = \sqrt{0^2 + 1^2 + 3^2} = \sqrt{10},$$

$$d_4 = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5},$$

$$d_5 = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2},$$

$$d_6 = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}.$$

So the distances are summarized below:

Obs.	Class	Distance to $(0, 0, 0)$
1	Red	3
2	Red	2
3	Red	$\sqrt{10}$
4	Green	$\sqrt{5}$
5	Green	$\sqrt{2}$
6	Red	$\sqrt{3}$

Ordering from nearest to farthest:

$$\sqrt{2} \text{ (Obs. 5, Green), } \sqrt{3} \text{ (Obs. 6, Red), } 2 \text{ (Obs. 2, Red), } \sqrt{5} \text{ (Obs. 4, Green), } 3 \text{ (Obs. 1, Red), } \sqrt{10} \text{ (Obs. 3, Red)}$$

2. **Prediction when $K = 1$:**

The single nearest neighbor is Observation 5, since

$$d_5 = \sqrt{2}$$

is the smallest distance.

Observation 5 has class Green, so the $K = 1$ prediction is

Green

3. **Prediction when $K = 3$:**

The three nearest neighbors are:

$$\text{Obs. 5 (Green), Obs. 6 (Red), Obs. 2 (Red).}$$

(a) **Uniform weights**

Each of the 3 neighbors gets weight $1/3$. Since among the 3 nearest neighbors we have

$$\text{Red: 2, Green: 1,}$$

the class probabilities are

$$P(\text{Red} \mid x_0) = \frac{2}{3}, \quad P(\text{Green} \mid x_0) = \frac{1}{3}.$$

Hence the prediction is

Red

.

(b) Distance weights

The weight formula is

$$w_i = \frac{e^{-d_i^2}}{\sum_{j=1}^K e^{-d_j^2}}.$$

For the 3 nearest neighbors:

$$d_5 = \sqrt{2}, \quad d_6 = \sqrt{3}, \quad d_2 = 2.$$

Therefore,

$$e^{-d_5^2} = e^{-2}, \quad e^{-d_6^2} = e^{-3}, \quad e^{-d_2^2} = e^{-4}.$$

The denominator is

$$e^{-2} + e^{-3} + e^{-4}.$$

So the normalized weights are

$$w_5 = \frac{e^{-2}}{e^{-2} + e^{-3} + e^{-4}} \approx 0.6652,$$

$$w_6 = \frac{e^{-3}}{e^{-2} + e^{-3} + e^{-4}} \approx 0.2447,$$

$$w_2 = \frac{e^{-4}}{e^{-2} + e^{-3} + e^{-4}} \approx 0.0900.$$

Since Observation 5 is Green and Observations 6 and 2 are Red:

$$P(\text{Green} \mid x_0) = w_5 \approx 0.6652,$$

$$P(\text{Red} \mid x_0) = w_6 + w_2 \approx 0.2447 + 0.0900 = 0.3348.$$

Thus under distance weighting, the prediction is

Green

.

Conclusion for $K = 3$:

Uniform weights: predict Red, with $P(\text{Red}) = \frac{2}{3}$, $P(\text{Green}) = \frac{1}{3}$,
Distance weights: predict Green, with $P(\text{Green}) \approx 0.6652$, $P(\text{Red}) \approx 0.3348$.

4. Scatter plot and decision boundary using only X_1 and X_2 :

Using only the first two features, the six observations become:

Obs.	X_1	X_2	Y
1	0	3	Red
2	2	0	Red
3	0	1	Red
4	0	1	Green
5	-1	0	Green
6	1	1	Red

Note that Observations 3 and 4 have the same $(X_1, X_2) = (0, 1)$ but different class labels, which is one of those delightful little data problems humans keep inventing.

The following Python code produces the scatter plot and the K -NN decision boundary (for example, with $K = 3$):

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.neighbors import KNeighborsClassifier

# Data using only X1 and X2
X = np.array([
    [0, 3],
    [2, 0],
    [0, 1],
    [0, 1],
    [-1, 0],
    [1, 1]
])

y = np.array(['Red', 'Red', 'Red', 'Green', 'Green', 'Red'])

# Encode labels for plotting
y_num = np.array([1 if label == 'Red' else 0 for label in y])

# Choose K
k = 3
knn = KNeighborsClassifier(n_neighbors=k)
knn.fit(X, y_num)

# Mesh grid
h = 0.02
x_min, x_max = X[:, 0].min() - 1, X[:, 0].max() + 1
y_min, y_max = X[:, 1].min() - 1, X[:, 1].max() + 1
xx, yy = np.meshgrid(np.arange(x_min, x_max, h),
```

```

np.arange(y_min, y_max, h))

Z = knn.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot decision boundary
plt.figure(figsize=(8,6))
plt.contourf(xx, yy, Z, alpha=0.3)

# Scatter points
for i in range(len(X)):
    if y[i] == 'Red':
        plt.scatter(X[i,0], X[i,1], marker='o', s=100, label='Red' if i == 0 else "")
    else:
        plt.scatter(X[i,0], X[i,1], marker='^', s=100, label='Green' if i == 3 else "")

plt.xlabel(r'$X_1$')
plt.ylabel(r'$X_2$')
plt.title(f'{k}-NN Decision Boundary using $X_1$ and $X_2$')
plt.legend()
plt.show()

```

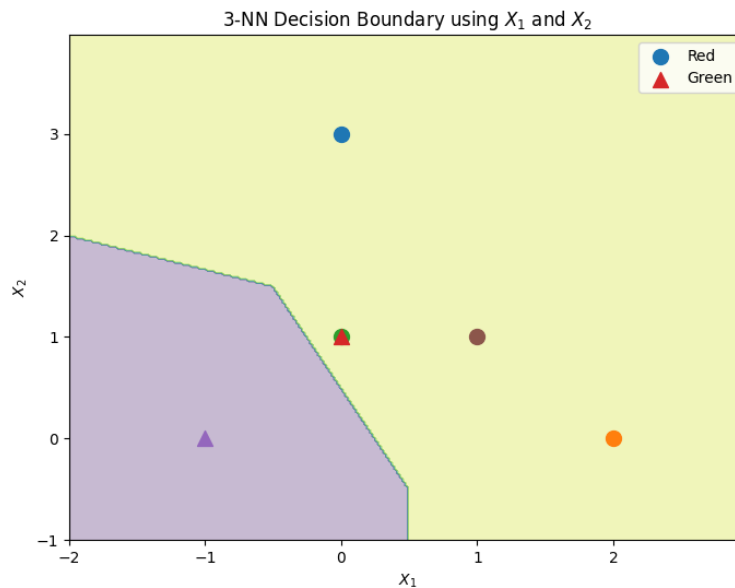


Figure 1: K-NN decision boundary using X_1 and X_2

Objective for k -means

Given the objective of k -means:

$$\min_{\mu_1, \mu_2, \dots, \mu_k} \sum_{i=1}^k \sum_{x \in \mathcal{C}_i} \|x - \mu_i\|_2^2. \quad (1)$$

Please prove that k -means algorithm will monotonically make the above objective non-increasing.

If we change the objective to:

$$\min_{\mu_1, \mu_2, \dots, \mu_k} \sum_{i=1}^k \sum_{x \in \mathcal{C}_i} \|x - \mu_i\|_1, \quad (2)$$

please design an algorithm (similar to k -means) to make the objective non-increasing and prove it.

Solution

Let the objective be

$$J = \sum_{i=1}^k \sum_{x \in \mathcal{C}_i} \|x - \mu_i\|_2^2.$$

(1) k -means is monotonically non-increasing

k -means alternates between two steps:

Assignment step: Fix $\mu_1, \dots, \mu_k$ and assign each point x to the nearest center:

$$x \in \mathcal{C}_i \text{ if } i = \arg \min_j \|x - \mu_j\|_2^2.$$

This minimizes each point's contribution, so the objective cannot increase.

Update step: Fix clusters $\mathcal{C}_1, \dots, \mathcal{C}_k$ and update

$$\mu_i = \frac{1}{|\mathcal{C}_i|} \sum_{x \in \mathcal{C}_i} x.$$

The mean minimizes $\sum_{x \in \mathcal{C}_i} \|x - \mu_i\|_2^2$, so the objective cannot increase.

Since both steps do not increase J , we have

$$J^{(t+1)} \leq J^{(t)}.$$

Thus, k -means is monotonically non-increasing.

(2) ℓ_1 version (k-medians)

New objective:

$$J_1 = \sum_{i=1}^k \sum_{x \in \mathcal{C}_i} \|x - \mu_i\|_1.$$

Algorithm (k-medians):

1. Assign each point to nearest center using ℓ_1 distance.
2. Update each center to the coordinate-wise median of its cluster.

Why it works:

- Assignment step: choosing nearest center minimizes each point's contribution $\Rightarrow$ objective does not increase.
- Update step: the median minimizes $\sum |x - \mu| \Rightarrow$ objective does not increase.

Therefore,

$$J_1^{(t+1)} \leq J_1^{(t)},$$

so the algorithm is also monotonically non-increasing.

Check the Notebook for k -means

Check the Notebook for K -NN